Visualization

Box Plot

A Box Plot is used to display the distribution of numerical data.

It shows:

- Minimum
- First Quartile (Q1)
- Median
- Third Quartile (Q3)
- Maximum
- Outliers

Scatter Plot

A Scatter Plot is used to show the relationship between two numerical variables.

It helps identify

- Positive correlation
- Negative correlation
- No correlation
- Outliers

Load Built-in Dataset

```
data(mtcars)

head(mtcars)
```

	mpg <dbl>	cyl <dbl>	disp <dbl>	hp <dbl>	drat <dbl>	wt <dbl>	qsec <dbl>	vs <dbl>	am <dbl>
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0

6 rows | 1-10 of 12 columns

Display Dataset Information

```
str(mtcars)
```

```
## 'data.frame':    32 obs. of  11 variables:
##  $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
##  $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
##  $ disp: num  160 160 108 258 360 ...
##  $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
##  $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
##  $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
##  $ qsec: num  16.5 17 18.6 19.4 17 ...
##  $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
##  $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
##  $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
##  $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

```
summary(mtcars)
```

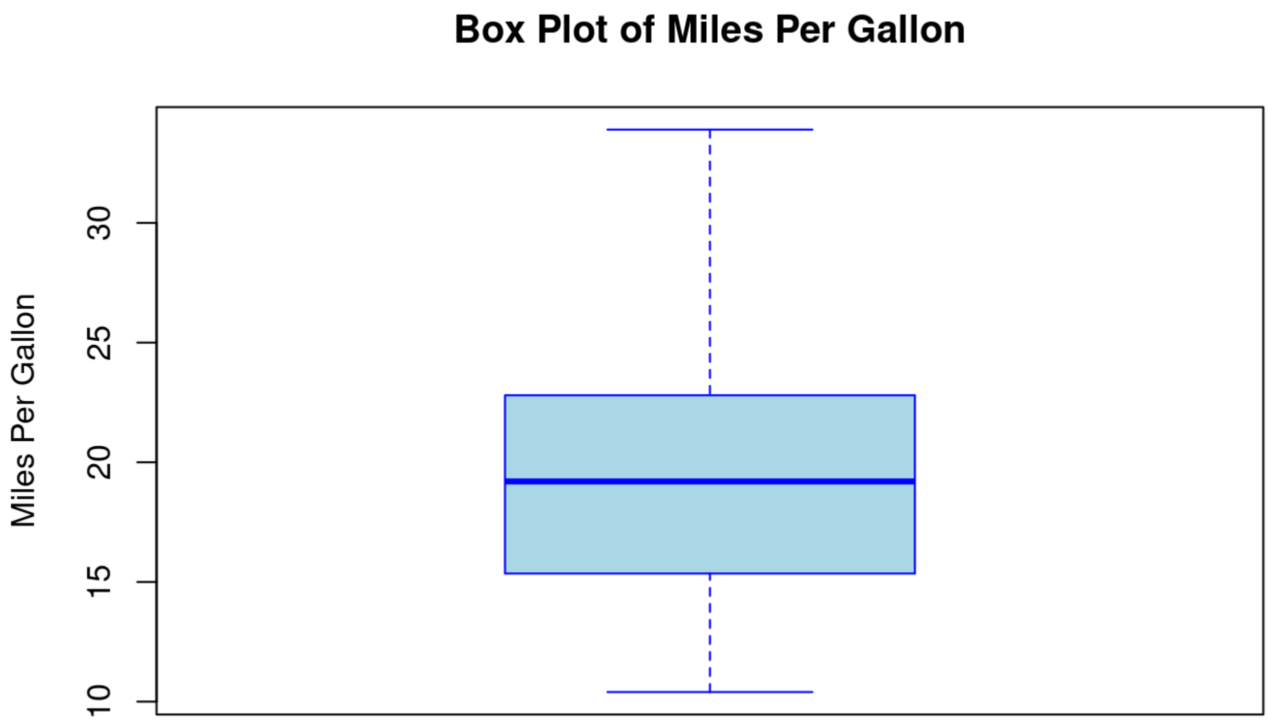
```
##       mpg             cyl             disp             hp       
##  Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
##  1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5  
##  Median :19.20   Median :6.000   Median :196.3   Median :123.0  
##  Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7  
##  3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0  
##  Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0  
##       drat             wt             qsec             vs        
##  Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
##  1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000  
##  Median :3.695   Median :3.325   Median :17.71   Median :0.0000  
##  Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375  
##  3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000  
##  Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000  
##        am              gear             carb      
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000  
##  Median :0.0000   Median :4.000   Median :2.000  
##  Mean   :0.4062   Mean   :3.688   Mean   :2.812  
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000  
##  Max.   :1.0000   Max.   :5.000   Max.   :8.000
```

```
dim(mtcars)
```

```
## [1] 32 11
```

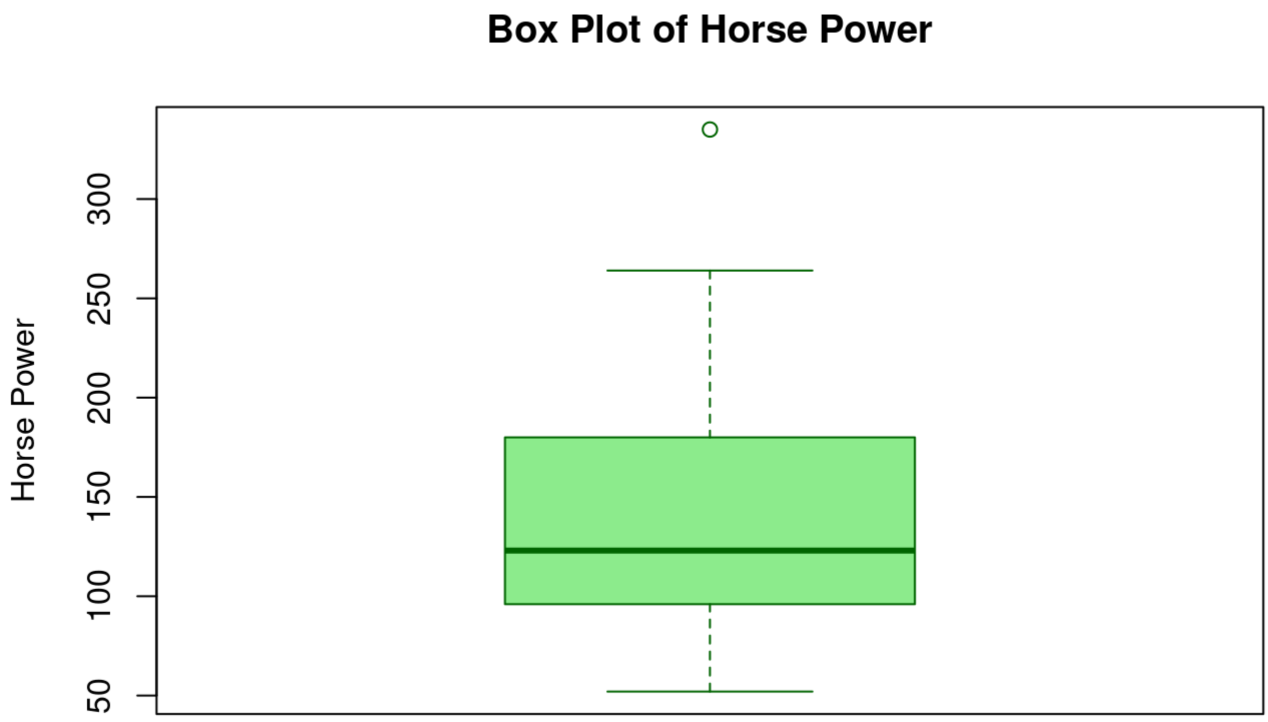
Box Plot of MPG

```
boxplot(
  mtcars$mpg,
  main="Box Plot of Miles Per Gallon",
  ylab="Miles Per Gallon",
  col="lightblue",
  border="blue"
)
```



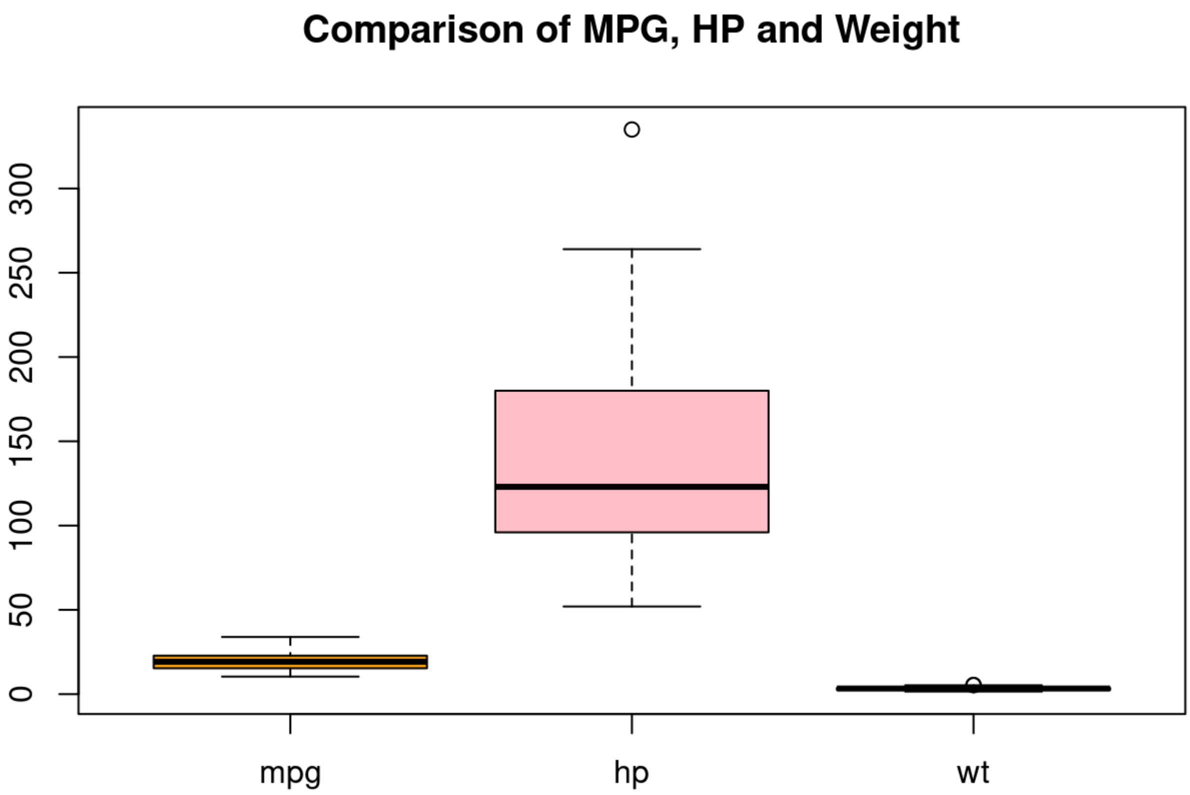
Box Plot of Horse Power

```
boxplot(
  mtcars$hp,
  main="Box Plot of Horse Power",
  ylab="Horse Power",
  col="lightgreen",
  border="darkgreen"
)
```



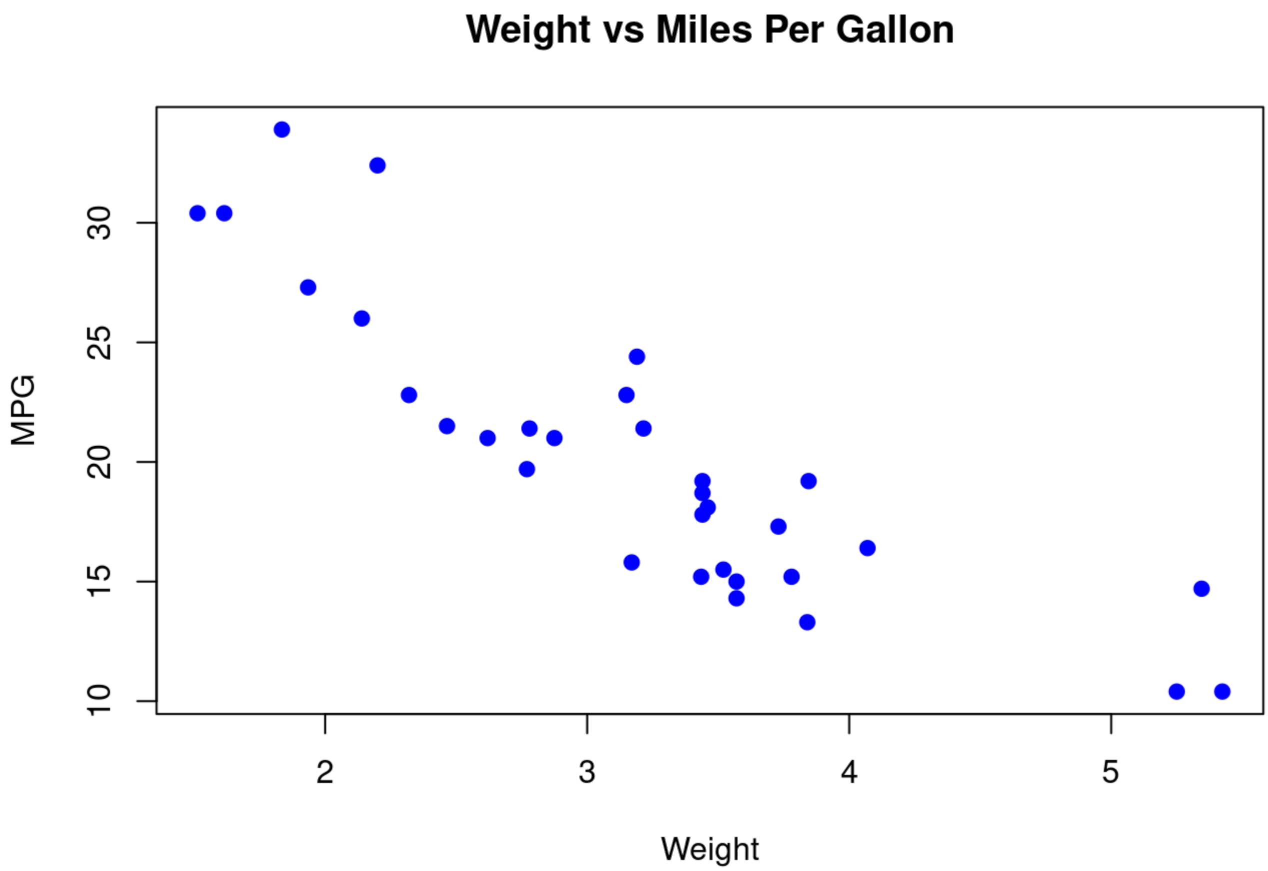
Box Plot of Multiple Variables

```
boxplot(
  mtcars[,c("mpg","hp","wt")],
  main="Comparison of MPG, HP and Weight",
  col=c("orange","pink","lightgreen")
)
```



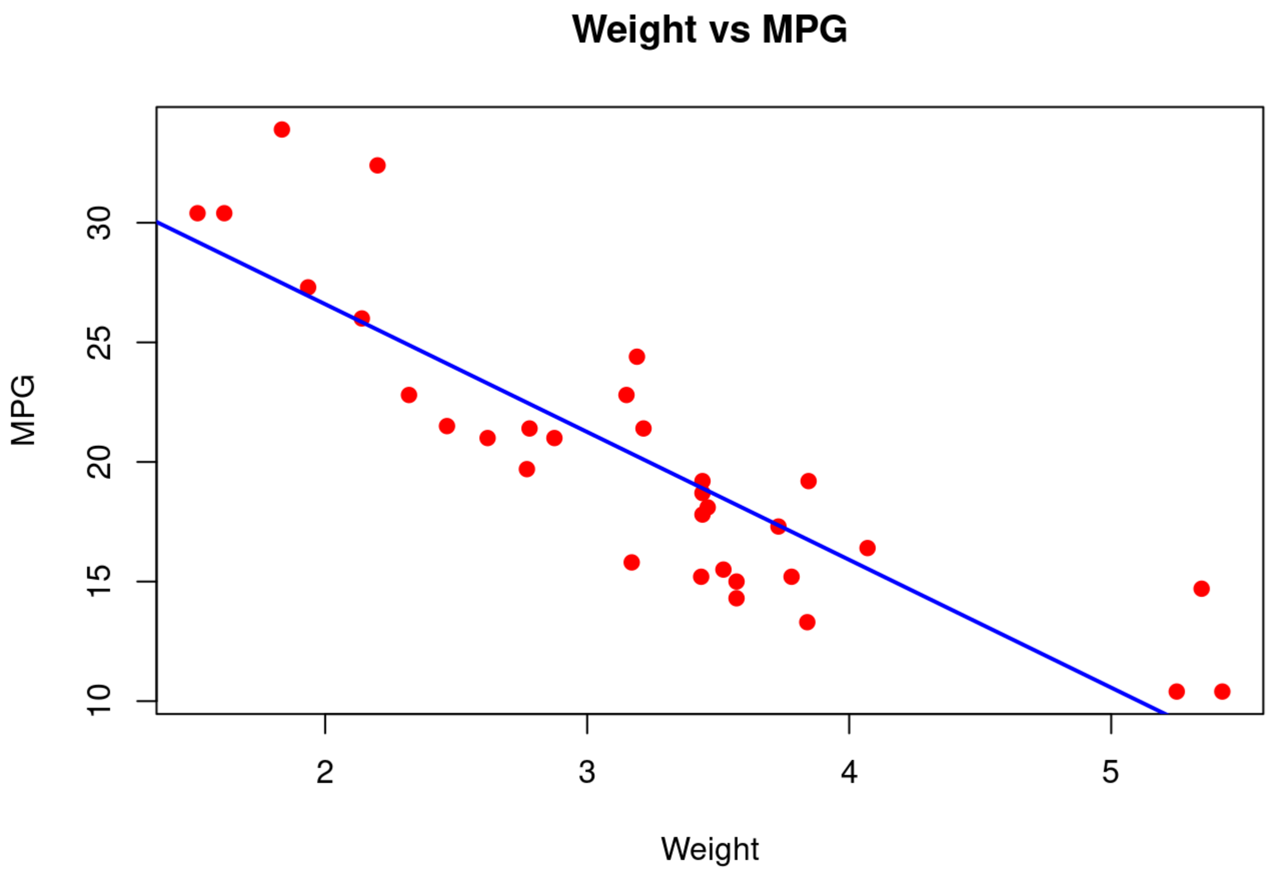
Scatter Plot (Weight vs MPG)

```
plot(
  mtcars$wt,
  mtcars$mpg,
  main="Weight vs Miles Per Gallon",
  xlab="Weight",
  ylab="MPG",
  pch=19,
  col="blue"
)
```



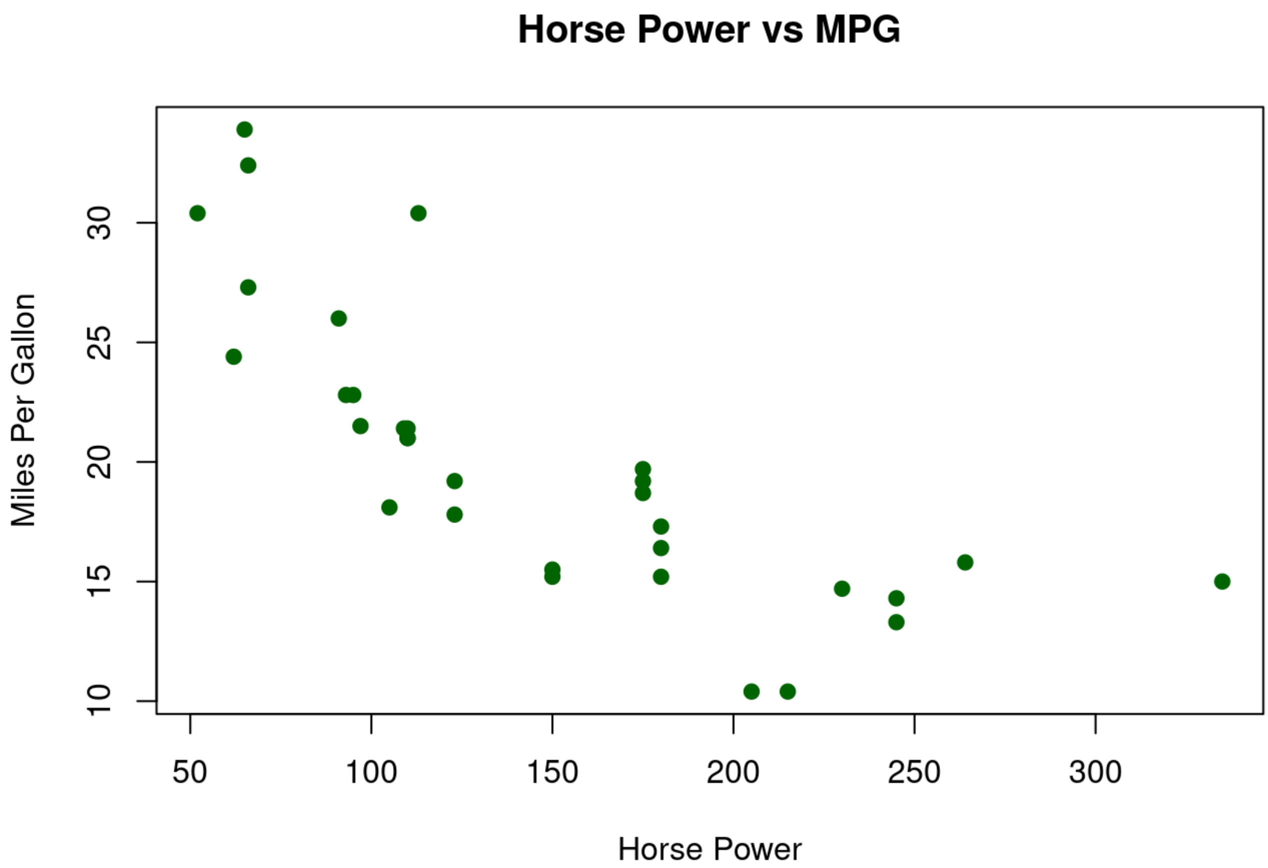
Scatter Plot with Regression Line

```
plot(
  mtcars$wt,
  mtcars$mpg,
  main="Weight vs MPG",
  xlab="Weight",
  ylab="MPG",
  pch=19,
  col="red"
)

abline(
  lm(mpg~wt,data=mtcars),
  col="blue",
  lwd=2
)
```



Scatter Plot (Horse Power vs MPG)

```
plot(
  mtcars$hp,
  mtcars$mpg,
  main="Horse Power vs MPG",
  xlab="Horse Power",
  ylab="Miles Per Gallon",
  pch=19,
  col="darkgreen"
)
```



Using Iris Dataset

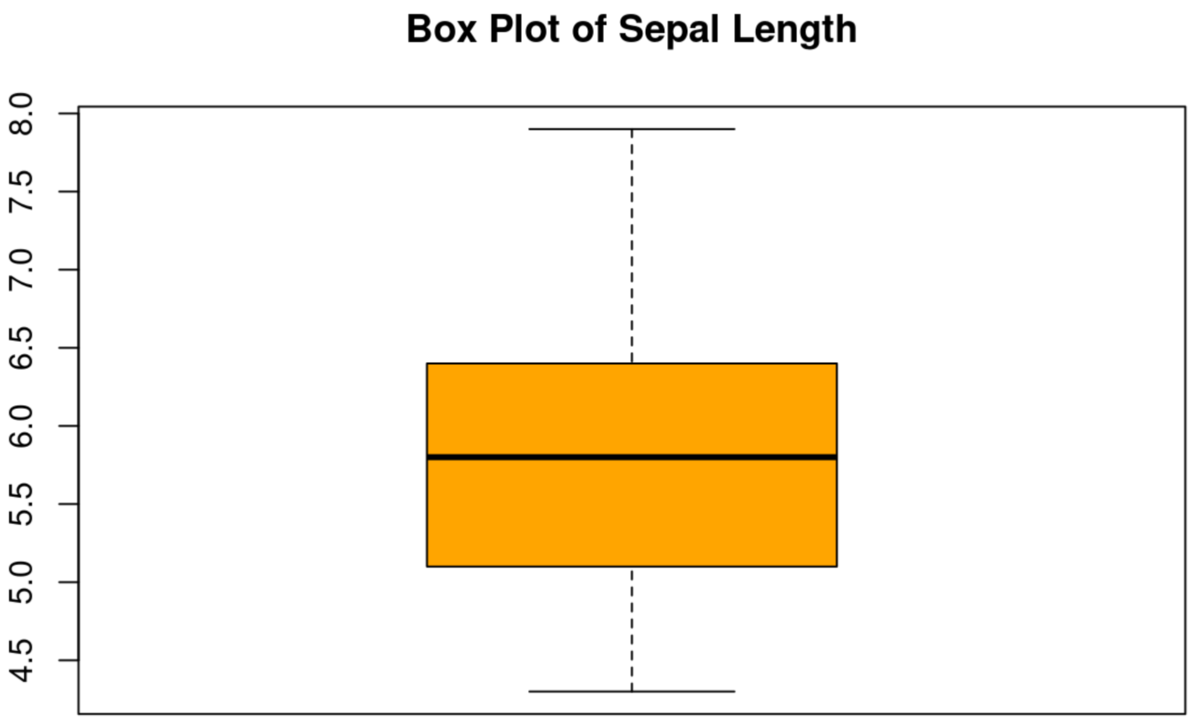
```
data(iris)

head(iris)
```

	Sepal.Length <dbl>	Sepal.Width <dbl>	Petal.Length <dbl>	Petal.Width <dbl>	Species <fct>
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

6 rows

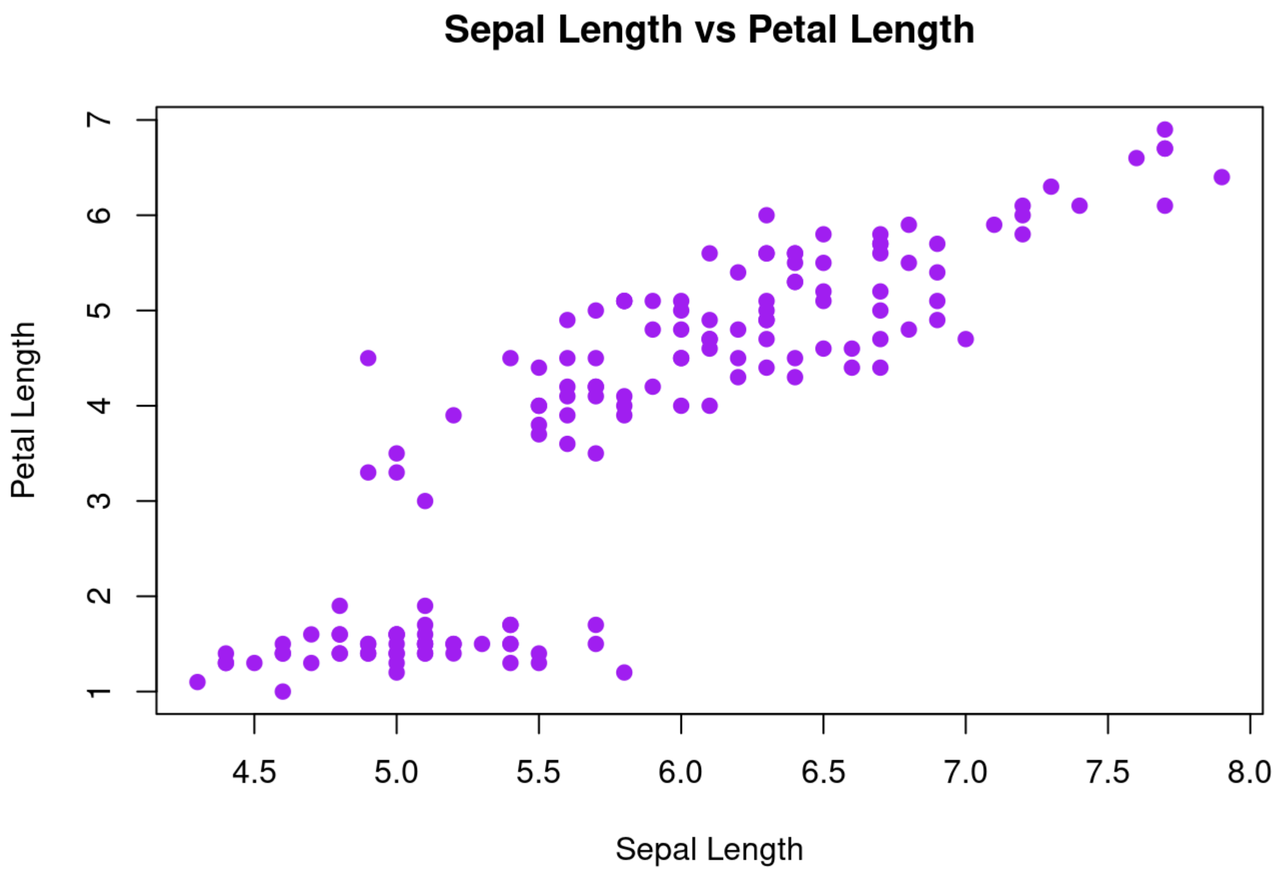
Box Plot of Sepal Length

```
boxplot(
  iris$Sepal.Length,
  main="Box Plot of Sepal Length",
  col="orange"
)
```



Scatter Plot of Sepal Length vs Petal Length

```
plot(
  iris$Sepal.Length,
  iris$Petal.Length,
  main="Sepal Length vs Petal Length",
  xlab="Sepal Length",
  ylab="Petal Length",
  pch=19,
  col="purple"
)
```


Sepal Length vs Petal Length
Petal Length
Sepal Length
1
2
3
4
5
6
7
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0